

BRIDGEPULSE · REQUIREMENTS

BridgePulse requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

BridgePulse requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with bridge owners or inspectors, and will be checked by calculation at TRL 3 and revised after co-design sessions (see BRP-PRB-001). Status is judged against the estimates in BRP-PRC-001; "met on estimate" is not a verification.

Reference installation. A steel or concrete footbridge or short-span road bridge of 5 to 30 m span, monitored at midspan, in a climate from -20 to +50 °C, with LoRaWAN coverage and no mains power. The worked example is the 7 m steel footbridge in BRP-PRC-001.

Table 1. Requirements and status at TRL 2.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Measure bridge acceleration	3 axes; noise density 25 $\mu\text{g}/\sqrt{\text{Hz}}$ or less; band 0.5 to 60 Hz	Datasheet; later bench noise test	Met on datasheet (22.5 $\mu\text{g}/\sqrt{\text{Hz}}$)
R2	Track natural frequencies	Every mode with a clear peak below 60 Hz, at least the first vertical mode; bin width 0.02 Hz or less per hourly record; repeatability 0.2 % (1σ) or better over a day of steady temperature	Spectral calculation; later field data	Bin width met by design (about 0.015 Hz); repeatability not verified
R3	Flag structural change	Flag a sustained shift of 1 % or more in a tracked frequency, after temperature compensation, within 14 days; one false flag per bridge per year or fewer	Analysis of public benchmark data (for example Z24), then field baseline	Not demonstrated; depends on excitation and the temperature model
R4	Measure strain	2 channels; range $\pm 1,000 \mu\epsilon$; resolution 2 $\mu\epsilon$ or better; drift 5 $\mu\epsilon$ per month or less after temperature compensation	ADC noise calculation; later outdoor drift test	Resolution plausible on estimate; drift not verified
R5	Measure temperature	Steel surface and shaded air; $\pm 0.5 \text{ }^\circ\text{C}$ or better, checked against a reference	Datasheet; CalRig chamber check	Not verified
R6	Stay within the FieldNode energy budget	Average draw 30 mW or less at the FieldNode sensor port	Power calculation; later current logging	Met on estimate (about 15 mW)
R7	Send and keep the data	One summary per hour of 48 bytes or less, within LoRaWAN regional limits at the data rate in use; raw records kept on site for 12 months or more	Payload design; storage calculation	Met on estimate (about 36 bytes; about 23 months on 32 GB); regional limits not checked
R8	Fit without harming the structure	No drilling, welding or cutting; strain gauges only on an area of paint removed and recoated with the owner's permission; fitted by two trained people in 4 h or less from below or beside the deck	Fitting sequence review; later trial fit	Met by design for drilling and welding; fitting time not verified
R9	Keep clear of what passes under and over the bridge	Nothing more than 15 mm below the girder soffit; nothing on the deck side except the FieldNode on a handrail post, with tamper-resistant fasteners	Model check	Met in the concept model (about 12 mm)
R10	Survive outdoors	Hub IP67, FieldNode IP65; operate from -20 to $+50 \text{ }^\circ\text{C}$; 5 years with one gauge recoat and no cell replacement	Datasheets and design review; later environmental checks	Not verified
R11	Protect privacy	No cameras or microphones; only structural readings leave the device	Design review	Met by design
R12	Affordable	Complete system, including the FieldNode core, \$250 or less in parts	Priced BOM (bom/ bom.csv)	Not met: about \$342 in total; about \$216 without the FieldNode core
R13				

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
	Present results responsibly	Every output labeled as monitoring data that supports inspection; no pass or fail safety rating anywhere in the system	Review of dashboard and report templates	Met by design intent; not yet implemented

■ Requirements not met or at risk

- **R12 (cost) not met:** about \$342 against \$250. Options are set out in `docs/REVIEW.md`.
- **R3 (change detection) not demonstrated:** it depends on quiet bridges giving a usable signal and on a temperature model that holds across seasons.
- **R2, R4, R5 and R10 not verified:** they rest on datasheets and estimates only.

■ Assumptions

- The accelerometer follows the ADXL355 datasheet summary ([Analog Devices](#)); a different part must meet R1.
- The FieldNode core offers about 115 mW average for sensors, as stated in its README, and a sealed M12 sensor port.
- A temperature-compensated frequency baseline needs at least four weeks of data before any flag, following the approach of [Peeters and De Roeck \(2001\)](#).
- Steel modulus 210 GPa for converting strain to stress.